Part 1 – Data Analytics Insights Report

Oct 16, 2026 · @Sook Jing Lee

Smart City Traffic Intelligence: Metro Interstate Traffic Volume (48,204 raw rows = 40,575 distinct hours, October 2012 to September 2018)

Summary

Traffic on this interstate follows the clock and the calendar far more than the weather. Time of day and day of week explain most of the variation; weather and temperature add only small effects. Heavy congestion (more than 5,500 vehicles per hour) occurs in about 15% of hours and is concentrated in the weekday rush hours.

Key findings

- **Rush hours dominate.** In 2017 the evening peak at 16:00 averages 5,834 vehicles per hour and the morning peak at 07:00 averages 4,820, against only 386 at 03:00.
- **Weekdays are busier than weekends.** Monday to Friday average 3,324 to 3,655 vehicles per hour; Saturday averages 2,816 and Sunday 2,458.
- **Weather has a modest effect.** Cloudy hours average 3,617 and clear hours 3,055. Fog (2,654), mist (2,882), snow (3,012) and thunderstorms (3,010) all sit below the overall average of 3,291, which suggests slower or fewer trips in poor visibility.
- **Temperature is a weak predictor.** The correlation between temperature and traffic is only 0.14, so temperature alone explains under 2% of the variation. Because temperature and traffic both follow the time of day, the correlation was also computed within each hour; it stays weak (0.05 to 0.18 for most hours, higher only at 19:00 and 20:00). Hours above 25°C are the exception: 22% to 29% of them are congested, against 10% to 17% in cooler bands.
- **Holidays are quiet.** On the five New Year's Day and Labor Day dates analysed (2014 and 2016 to 2017 for New Year's Day, 2015 to 2017 for Labor Day), whole-day traffic averages 1,637 to 2,430 vehicles per hour, about 25% to 50% below the year's average. Across all 11 holidays the whole-day average runs from 1,764 (Christmas Day) to 3,837 (State Fair); eight are clearly below the overall average of 3,291, while Veterans Day, Columbus Day and State Fair are at or above it.
- **Weather and congestion are only loosely linked.** Congestion is somewhat less likely in clear weather (odds ratio 0.74), but the association is modest and partly reflects when each weather type occurs.

Implications for the mobility team

- Signal timing, lane management and incident response matter most at 07:00 and 15:00 to 17:00 on weekdays.
- Weather alerts alone will not predict congestion. Time features (hour, weekday, holiday) should be the core of any forecasting model, with weather as a secondary input.
- Hot afternoons deserve extra attention, since congestion is noticeably more common above 25°C.

- Overnight and weekend capacity is under-used, so roadworks are best scheduled then.

Data quality issues

- **One row per hour.** The 48,204 rows cover only 40,575 distinct hours, because the same hour is repeated once per weather description with identical traffic volume. Left alone, every average, correlation and probability would be weighted by weather labels. All statistics here use one row per hour (the most severe weather label is kept), applied in SQL as the view `traffic_hourly`.
- **Nine missing months.** There are no records from September 2014 to May 2015, 2012 starts in October and 2018 ends in September. Yearly totals therefore measure coverage, not traffic (2,103 hourly rows in 2012 against 8,713 in 2017), so compare yearly averages (3,194 to 3,377 per hour), not totals. The apparent 2014 to 2015 fall is a coverage effect.
- **A required year does not exist.** New Year's Day 2015 falls in the missing months, so it is reported as no data and 2014 is used as the substitute.
- **Holiday flag covers one hour only.** All 61 flagged rows sit at 00:00 and cover 53 holiday dates (8 rows repeat the same hour), so the raw flag marks one hour per holiday. It was filled across the full day before holiday traffic was measured, giving 1,203 holiday hours (53 dates × 24 = 1,272 possible, less 69 hours missing from the data on 17 of those days).
- **Impossible values.** 10 temperature readings of 0 K (−273°C) on 31 January and 2 February 2014, a rainfall reading of 9,831.3 mm in one hour, and zero traffic at 18:00 and 23:00 are sensor errors and should be imputed with monthly medians.
- **Anomaly.** 23 July 2016 averages only 277 vehicles per hour across 24 readings and needs checking against road-closure or sensor logs. Part 2 later showed the low-traffic stretch is wider than one day: from 22:00 on 22 July to about 17:00 on 24 July (44 hours, 28 of them under 100 vehicles*).*
- **Rare categories.** Smoke has only 2 hours and Squall only 1, so their averages (4,443 and 420) are unreliable.
- **Codes.** The text "None" in the holiday column means no holiday, not a missing value.
- **Three congestion definitions.** A fixed threshold above 5,500 vehicles per hour (policy-facing, 15.1% of hours), the Low/Medium/High dashboard bands (below 4,500, 4,500 to 5,500, above 5,500) and quartiles (1,248 / 3,427 / 4,952, used later as a modelling construct) are kept distinct.

Recommendations

1. Keep one row per hour in every part of the project, and clean the impossible values (0 K, 9,831 mm, zero traffic) before modelling.
2. Build forecasting features from hour, weekday, holiday (filled across the whole day) and a rush-hour flag first, then add weather.
3. Investigate the 23 July 2016 event with the data owner.
4. Report congestion as a share of hours above 5,500 vehicles, tracked by hour and weekday on the dashboard, and state the coverage gaps alongside any yearly comparison.

Dashboard screenshot:

